

Hidden Markov Model for POS Tagging

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1 Introduction

This work uses the Wall Street Journal (WSJ) portion of the Penn Treebank dataset to develop a Part-of-Speech (POS) tagger for English based on a Hidden Markov Model (HMM) with Viterbi decoding. The goal is to decode the most likely tag sequence for unknown phrases and estimate transition and emission probabilities from trained data.

2 Dataset and Experimental Setup

We use the WSJ portion of the Penn Treebank corpus, split according to the standard convention:

- **Training set:** Sections 00–18
- **Development set:** Sections 19–21
- **Test set:** Sections 22–24

Each sentence consists of tokens in the format `word/TAG`. The dataset includes 36 core POS tags (45 when punctuation is included).

3 Baseline Model

3.1 Baseline Performance Summary (NLTK Tagger)

Metric	Value
Overall Accuracy	0.97
Weighted F1 Score	0.97
Macro F1 Score	0.86–0.87

Table 1: Baseline NLTK POS Tagger Performance

3.2 Observations

- High-frequency tags such as NN, DT, IN, and NNP achieve very high precision and recall.
- Rare tags (e.g., LS, PDT, RBS) suffer from undefined or zero precision due to extremely low support.

These results confirm that the evaluation pipeline and data splits are functioning correctly. This baseline will later be used to directly compare against the HMM-based POS tagger.

4 HMM Model (Still in Progress)

A bigram Hidden Markov Model is currently under active development with the following design:

4.1 Transition Probabilities

$$P(t_i | t_{i-1})$$

4.2 Emission Probabilities

$$P(w_i | t_i)$$

4.3 Add- α Smoothing

Currently set to $\alpha = 1.0$.

4.4 Decoding Algorithm

Standard Viterbi algorithm in log-space to avoid numerical underflow.

4.5 Special Tokens

<START>, <STOP>, and <UNK> are used to handle sentence boundaries and unknown words.

5 Planned Work (Still in Planning)

- Hyperparameter tuning — In progress
- Confusion matrix and per-tag analysis — In progress
- Final accuracy comparison — In progress

6 Remarks

This report documents the successful setup of the POS tagging pipeline and establishment of a strong baseline using the NLTK tagger. The custom HMM with Viterbi decoding is under active development and validation. Full experimental results, optimization, and error analysis will be included in the final submission.